

VIVEK RAJ SINGH

Bhopal, Madhya Pradesh

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EDUCATION

Vellore Institute of Technology

B.Tech - CSE (Cloud Computing) – CGPA: 8.69

2023 – 2027

Bhopal, M.P.

QuantInsti

Executive Programme in Algorithmic Trading (EPAT)

Sep 2025 – May 2026

Online

EXPERIENCE

Quintal Mind

Quantitative Research Intern

May 2026 – Present

Remote

- Optimized backtesting infrastructure in Python to process 500K+ high-frequency option chains over 3 years, evaluating Cross Butterfly and Time-based Straddle strategies.
- Developed mean-reversion signals by isolating deviations between live prices and theoretical curves, achieving a 1.8 Sharpe ratio in out-of-sample testing.
- Built an event-driven, minute-level trading simulator with dynamic risk management, trailing stops, and volatility regime-change exits.

FOSSEE, IIT Bombay

Summer Research Fellow (Part-Time) – Mathematical Modeling & Quantitative Finance

May 2026 – Present

Remote

- Achieved ~65% variance reduction in path dependency by engineering a Heston Stochastic Volatility engine using Euler-Maruyama Monte Carlo paths and Antithetic Variates; published on Scilab portal.
- Extracted Black-Scholes implied volatilities using bisection root-finding to plot volatility smiles and rendered 3D option surfaces over a 13x10 grid.

TECHNICAL SKILLS

- Languages: Python, C++, SQL, Rust, Scilab
- Quant & ML: PyTorch, NumPy, SciPy, pandas, statsmodels, scikit-learn, TensorFlow
- Tools: Git, Docker, FastAPI
- Quantitative Concepts: Monte Carlo, Options Pricing, Stochastic Calculus, VaR/CVaR, Time-Series Analysis

PROJECTS

Stochastic Risk Quantification Engine 🌀 | Live Demo | PyTorch, statsmodels, Monte Carlo, FastAPI

- Modeled storage degradation as a stochastic process, developing a forecasting ensemble combining Deep Learning (LSTM with Cosine Annealing, AdamW) and statistical ARIMA models.
- Engineered a risk engine running 10,000 Monte Carlo simulations to compute VaR_{95} and $CVaR_{95}$ bounds, implementing Isolation Forests (3σ thresholding) for threat classification.
- Exposed model inference via a thread-safe FastAPI service with serialized model bundles for real-time risk scoring.

Mean-Variance Portfolio Optimizer 🌀 | Python, NumPy, cvxpy, Quantitative Modeling, Monte Carlo Simulation

- Built a Markowitz Mean-Variance optimizer in Python using NumPy and cvxpy, modeling a multi-asset portfolio across 50 weight combinations to plot the Efficient Frontier.
- Simulated 25 independent price paths over 20 trading days calibrated to historical volatility to estimate forward price dispersion and Value at Risk (VaR) with $\pm 2\sigma$ Bollinger Bands.

ROCm-Forge: CUDA-to-ROCm AI Copilot 🌀 | Live Demo | Python, C++, CUDA, LLMs, Static Analysis

- Featured in AMD Developer Hackathon 2026: Engineered a 9-agent autonomous migration engine translating NVIDIA CUDA to AMD ROCm, reducing manual effort by 65%.
- Developed static analysis pipelines using Python AST parsing and a hardware-aware scanner to detect warp-to-wavefront (32 to 64 thread) mismatches and Tensor Core lowerings.
- Fine-tuned CodeLlama-7B via QLoRA on AMD Instinct MI300X GPUs (ROCm 6.2) using a curated dataset of 20+ paired CUDA-to-ROCm migration examples.